Choice-Set Restriction in Machines and People

Independence of Irrelevant Alternatives fails in both,
and far more sharply in machines

Peter Cotton

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Abstract

Whether choice probabilities obey Luce's Choice Axiom has been argued since Thurstone proposed the alternative in 1927, and it was never settled on evidence, because a human choice distribution must be estimated from repeated trials while the two families differ by little on any single menu. Language models report the distribution directly. We use them as an instrument and compare the result with archival human rankings. Naming a choice down to two items should leave their odds untouched; instead both parties discard much of their prior odds. Fitting the discount $\delta = -\lambda \log(p_i/p_j)$, where $\lambda = 0$ is the axiom and $\lambda = 1$ abandons the prior ranking entirely, gives 0.42 for human preferences and 0.19 for human perceptual judgments, against 0.69 $[0.585, 0.792]$ for machines. A Thurstonian contest calibrated to each population predicts 0.32, 0.16 and 0.12, so people exceed the contest by a fifth to a third while machines exceed it almost six-fold: people are close to the race and far from the axiom, machines close to neither. Machine choice also fails in a way no random utility model permits, reversing a third of surviving pairs.

1 Introduction

Large language models can be interrogated cheaply and at scale, which makes the study of their revealed preferences attractive whether one regards them as incipient minds or as statistical mimics of our own predilections. Either way, a question from the psychology of choice arises immediately: when a model distributes probability over alternatives, what law governs how that distribution changes as the set of alternatives changes?

The workhorse answer in economics and psychology is Luce's Choice Axiom [22], which asserts that the probability of selecting item i from a set S is proportional to a fixed utility $u(i)$:

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}. \tag{1}$$

Its signature is Independence of Irrelevant Alternatives: the ratio $P(A)/P(B)$ is unchanged by the presence or absence of a third option C.

There is a natural alignment between this assumption and the token probabilities of language models, which invariably emerge from a softmax:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}, \tag{2}$$

however complicated the transformer computation producing z_i may be. Delete item 1 by setting $z_1 = -\infty$, a surgical procedure impossible in humans, and the remaining probabilities are the old ones renormalized by $1/(1-p_1)$. The output layer therefore satisfies the Choice Axiom exactly, for that operation.

Restricting a choice set in language is not that operation, and we do not suppose it is. Telling a model that an option is unavailable changes the prompt, so the whole logit vector is recomputed rather than masked, and nothing in the architecture constrains the new odds among survivors to match the old ones. Menu-dependent logits can represent any positive choice rule at all. The softmax is thus a statement about what the sampling head does with a *fixed* state, not a prediction about behaviour across prompts, and the question of which law describes the recomputation is empirical from the outset.

That gap is the subject of this paper, and it has practical bite because masked renormalization is exactly what deployed systems compute: constrained decoding, tool routers that mask unavailable tools, and guardrails that drop a candidate all impose Equation (1) by construction. What a model reports when told in context that an option is unavailable is a different object. We measure how far apart the two are, and find the difference better described by the Thurstonian contest of [24] than by renormalization.

The evidence is not a single experiment. Table 1 lists thirteen designs comprising 73,704 scored choice cells, built from 61,512 logged exact log-probability measurements on five commercial and fourteen open-weight models, alongside the masked language model study of the 2024 version and a generative replication on Claude. Every cell is a zero-parameter out-of-sample prediction: both families are calibrated to an unrestricted distribution and asked to predict a restricted one they have not seen. The two largest batteries, 3,660 and 1,195 cells, favor the contest model with bootstrap confidence intervals excluding zero on every one of the five models, and the margin rises monotonically with model tier, from +0.30 on GPT-4.1-nano to +0.88 on GPT-4.1. All raw model responses are logged and released, so every number below is recomputable without further inference.

Section 2 states the two competing models and Section 2.1 explains why one of them won an argument it did not settle. Section 3 puts the axiom to its own test and rejects it. Section 4 shows that the failure is a reordering rather than a reweighting, which rules out both families at once and reframes everything after it as a question about approximation. Sections 5 and 6 give the masked language model evidence from the 2024 version, and Section 8 onward asks which rule forecasts best, in what regime, and how the answer varies with model tier, post-training, elicitation depth and menu structure. Section 9 draws out the implications, and separates what the measurements support from three conjectures they merely motivate: a mechanism for generative "mode collapse", a diversity knob, and a contest-structured output layer.

2 A choice of choice models

One might describe Equation (1) as an *urn model*: item k is represented by balls in an urn in proportion to $u(k)$, a choice is a random draw, and shrinking the choice set merely removes balls. The renormalization of probability under restriction then seems perfectly reasonable.

But perhaps preference is a horse race, not an urn. In the competing model each item i carries a location parameter a_i (a dislikability, say) and an associated performance $X_i \sim N(a_i, 1)$. Item i is chosen when its performance is the lowest draw. This is the model of Thurstone [24], and it makes different predictions about how choice probabilities respond when the choice set is

Design	Models	Cells	Verdict
Adjective qualification (masked)	2 masked LMs	100+	Thurstone $\approx 80\%$
Two-slot elimination (masked)	2 masked LMs	100+	Thurstone $\approx 80\%$
Polysemous generation, restriction	Claude	21	re-collapse, not renormalization
Qualified-noun restriction	3 GPT	71	tie (degenerate priors)
Random elicitation, one deletion	3 GPT	77	Thurstone 39/56, $p = 0.002$
Permutation-controlled deletion	5 GPT	1,195	+0.025 [+0.009, +0.047]
Original 2024 stimuli, two-slot	5 GPT	3,660	+0.519 [+0.443, +0.600]
Open-vocabulary sweep, 99 question types	3 GPT	37,764	**99/99 types favor Thurstone**
pooled estimate			+0.150 [+0.135, +0.165]
Ordered selection to depth five	5 GPT	1,363	+0.339 [+0.248, +0.438]
Reversibility (best vs worst first)	5 GPT	108	−0.290 [−1.060, +0.425]; neither family
Block–Marschak / RUM test	5 GPT	30	30/30 violate RUM
Menu deletion under conditioning	5 GPT	229	+0.026 [+0.008, +0.045]
Duplicates, decoys, compromise	5 GPT	101	substitution 15/41; regularity 23/60
Local panel on the same stimuli	14 open	24,948	tuning effect in 4 of 6 pairs
Local inventory panel	14 open	4,158	+0.015 [+0.012, +0.018]

Table 1: The full evidence base, recomputed from the released logs. Cells are scored zero-parameter predictions of a restricted distribution, and intervals are cluster bootstraps over the design's natural grouping (question type, category or family) rather than over cells. Every design but the polysemy replication uses exact log probabilities. Masked language model counts are category/adjective combinations from the 2024 version.

altered: rather than removing a ball from the urn, we remove a horse from the race.

Unit variance and independence make this Case V, the family's default and its tightest special case. Everything below uses it, and Section 8.7 explains why: richer contests would fit better, but Luce in its bare form has nothing to fit, so comparing a searched Thurstone against an unsearched Luce would prove little.

(As an aside, it *is* possible to engineer a contest that satisfies Luce's Choice Axiom, by taking $X_i \sim \mathrm{Exp}(h_i)$ with common location and differing hazard rates, or, by Yellott's theorem, with Gumbel noise [33]. The comparison here can therefore be viewed as between two noise families within the contest framework, which is why single-menu fits cannot separate them and the restriction, duplication, and context designs below are required.)

The practical obstacle to Thurstonian models has always been computation, not plausibility. A fast lattice algorithm for calibrating multi-entrant Thurstone models, recovering locations $\{a_i\}$ from winning probabilities and computing derived quantities for fields exceeding 100,000 entrants, was given in [18], and is what makes the experiments below routine.

2.1 A hundred years of the same argument

The question this paper puts to language models is not a new one. It has been contested continuously since Thurstone published the law of comparative judgment in 1927 [24], building on Fechner's discriminal dispersions [1], and it has never been settled in either direction.

The interesting historical question is not who was right but why one side won, because it did not win on the evidence. It won on arithmetic.

Thurstone's contest has no closed form beyond two alternatives. Choosing among n items requires the probability that one normal draw is extremal among n, an $(n-1)$-dimensional integral

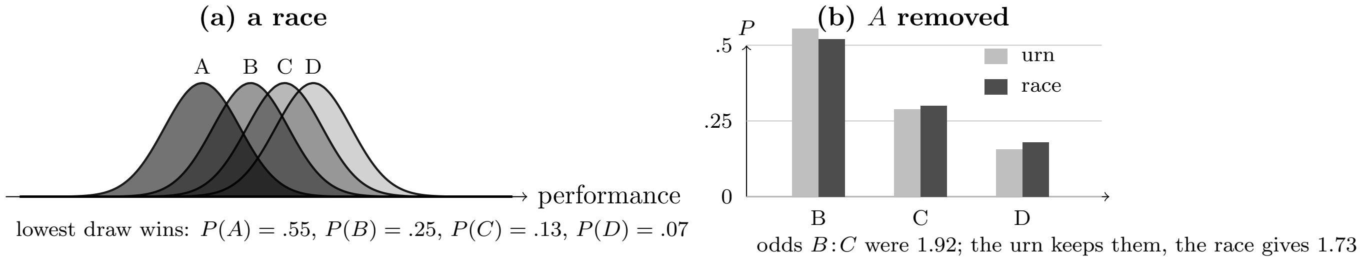

Figure 1: The two models disagree only under restriction. In (a) each item draws a performance and the lowest wins, which for these locations gives the win probabilities shown. In (b) item A is removed. The urn rescales what is left, so the odds between any two survivors are unchanged by construction. The race is re-run without A, and the mass it vacated is redistributed unevenly: B gains less than proportionally and the tail gains more, moving the $B:C$ odds from 1.92 to 1.73. Every battery in this paper measures which of these two reallocations a language model performs. Locations are calibrated to the win probabilities in (a) by the method of [18]; the numbers are exact, not illustrative.

with no elementary expression. Case V, the equal-variance independent case that Mosteller [2] made usable, reduces the *paired* comparison to $\Phi(d/\sqrt{2})$ and no further. Everything past a pair needed numerical integration on machinery that did not exist, and when the econometric version was finally written down as multinomial probit [8] it was reported as practical only for a handful of alternatives. Simulation methods made it estimable in the 1990s and it remains the expensive option [9].

The logistic alternative had the opposite property. Bradley and Terry [3] wrote a paired comparison model with a closed form, a concave log-likelihood and estimation by hand. Luce [22] then supplied the axiom, deriving Equation (1) from Independence of Irrelevant Alternatives, which converted a computational convenience into a behavioral principle: the form was no longer merely tractable, it was *implied* by a plausible-sounding condition on choice. McFadden [11] made it the workhorse of applied discrete choice, and Yellott [33] completed the retrospective justification by showing exactly which noise distribution the axiom corresponds to, which had the side effect of making the double exponential look canonical rather than merely convenient.

The pattern of the century that followed is that the Luce form was repaired rather than replaced. Nested logit, the generalized extreme value family and mixed logit [16] were all built to relax the independence IIA imposes while keeping the closed form that made the logit estimable. Choosing the Thurstonian route instead would have meant paying the integral, and almost nobody did.

The objections were immediate and they were about restriction. Debreu [28], reviewing Luce's book, observed that adding a near duplicate of an existing alternative should divide that alternative's share rather than draw proportionally from everything, which IIA cannot do. Tversky [10] showed that similarity between alternatives violates simple scalability altogether and proposed elimination by aspects instead. Luce [12] himself reviewed the axiom's standing after twenty years and treated its empirical failures as established rather than contested. In econometrics the same problem drove a long sequence of repairs: McFadden's conditional logit [11] made the Luce form the workhorse of discrete choice, then nested logit, the generalized extreme value family and mixed logit [16] were built to relax the independence it imposes, with Hausman and McFadden [14] providing a specification test for the violation.

Two results from that literature bear directly on what follows. Yellott [33] proved that Luce's axiom holds in a random utility contest if and only if the noise is double exponential, which places the two families inside one framework and identifies the disagreement as a question about noise rather than about utility. Block and Marschak [31] and Falmagne [32] characterized when a complete choice structure admits any random utility representation at all, by nonnegativity of the Block–Marschak polynomials, which supplies a test that does not presuppose either family. Section 8.11 applies it.

The dispute also has a quantitative history outside psychology. In racing markets, Harville's renormalization [20] is the Luce prediction applied to finishing order, and its systematic mispricing of second and third place is well documented; Henery [13] and Stern [15] proposed normal, that is Thurstonian, order models to correct it, and the correction works. The two-slot design of Section 5 is that comparison transplanted into a language model.

Two things have changed, and together they are why the question is worth reopening now rather than in 1977. The integral is no longer expensive: the lattice method of [18] returns the whole vector of winning probabilities for fields of 10^5 entrants, so the tractability argument that decided the matter has expired. And the measurement is no longer noisy. A century of this dispute was conducted on human subjects, where a choice distribution must be estimated from repeated trials and the standard errors swamp the difference between the two families on any single menu. A language model reports its distribution directly, to as many digits as one wants, which turns a question that needed thousands of participants into one that needs a few thousand API calls.

What makes the machine case worth revisiting is also that one side of this argument has been built into the apparatus. A softmax output layer is the Luce form, so an LLM's sampling head satisfies the Choice Axiom by construction. Reward models for preference tuning are Bradley–Terry [3], fitted to pairwise human comparisons on the assumption that a single scalar utility governs choice, and Direct Preference Optimization [17] inherits the same likelihood. Listwise ranking objectives use Plackett–Luce [23]. The modern pipeline therefore assumes the Luce side of a century-old and unresolved dispute at three separate points, which makes the question of whether the resulting behavior obeys the axiom an empirical matter with some consequence, and one that exact log probabilities can settle in a way that human subjects never could.

2.2 Why ask a machine

Whether a language model's behaviour is worth studying with the methods of psychology is a question others have already settled in the affirmative, and we take it as settled rather than argue it again. Binz and Schulz [4] ran a battery of canonical cognitive experiments on GPT-3 and found human-like performance alongside human-like failure. Argyle and colleagues [5] showed that conditioning a model on demographic detail reproduces response distributions of the corresponding human subgroups closely enough to be useful as samples. Dillion and colleagues [6] set out when such substitution is and is not defensible, and a programme has since been named for the activity [7]. The stated-versus-generated gap that motivates Section 8 is itself a finding of that literature [25].

Our reason for asking a machine is narrower and methodological. A century of this dispute was conducted where the quantity in question cannot be observed. A human choice distribution has to be estimated from repeated trials, one categorical outcome at a time, and the difference between the two families on a single menu is small enough that the standard errors swallow it.

That is why the argument was settled by tractability rather than by measurement. A language model reports the distribution itself, to as many digits as one cares to read, and it will answer the same question under a hundred variations of the choice set without fatigue or practice effects. The dispute becomes a measurement problem rather than a statistical one.

Two caveats belong here rather than at the end. Nothing in this paper is evidence about human choice: similarity of behaviour would not establish similarity of mechanism, and we have no human data. And the instrument brings its own artefacts, of which the largest we found is position: merely swapping which of two options is named first moves their log odds by around 1.4 nats (Section 3), comfortably more than the effect any choice model here predicts. Machine subjects trade sampling noise for prompt sensitivity. They are not quieter, they are noisy in a different and more controllable way.

3 The axiom's own test

Every battery in the older literature, and most of those below, scores a whole restricted distribution. That entangles three things: where the mass goes, how spread out it becomes, and how many alternatives survive. The axiom itself entangles none of them. Luce's condition is that for any two items

$$\frac{P(i)}{P(j)} \quad \text{does not depend on what else is available,} \tag{3}$$

so naming a choice down to exactly $\{i, j\}$ is the sharpest available probe. The statistic is a shift in log odds,

$$\delta_{ij} = \log \frac{q_i}{q_j} - \log \frac{p_i}{p_j}, \tag{4}$$

where p is unrestricted and q is measured with the set named down to the pair. Being a ratio of ratios, δ is invariant to any rescaling of either distribution, so none of the flattening that dominates Section 8.7 can enter, and no temperature or noise scale can be fitted to absorb it. The axiom's prediction is not approximate. It is $\delta_{ij} = 0$.

For each of 50 categories and five models we read the unrestricted distribution, then name the choice down to a pair, then to that pair plus one low-probability distractor, for pairs spanning the odds range. Listing order is crossed: every pair is asked in both orders, because it turns out to matter more than anything either choice model predicts.

Condition	Shift in log odds δ
Favourite named first	-1.784 $[-2.193, -1.384]$
Favourite named second	-0.516 $[-1.015, -0.059]$
Order-averaged	$\mathbf{-1.150}$ $[\mathbf{-1.547}, \mathbf{-0.808}]$
Pair plus one distractor	-4.017 $[-4.467, -3.543]$
Contest (Case V) prediction	-0.259 $[-0.311, -0.211]$
Choice Axiom prediction	0 exactly

Table 2: Odds invariance on 279 pairs, 50 categories, five models, clustered by category. Naming a choice down to two items shrinks the favourite's odds by roughly a factor of three. The Choice Axiom requires no change at all. The contest predicts the direction and about a quarter of the magnitude.

Three things follow. First, the axiom is rejected in essentially every pair, not merely on average: the order-averaged shift exceeds 0.1 nats in 96.8% of pairs and 0.5 nats in 80.3%, against a prediction of exact invariance. Restricting a language model's choice set does not preserve the relative standing of the survivors, and the mean shift of -1.15 nats is a factor of 3.2 in odds.

Second, the direction of the failure is the contest's, though not universally. Case V predicts shrinkage toward parity in 276 of 279 pairs, because re-running a race over a shorter field compresses the favourite's advantage, where an urn predicts no movement at all. The odds did shrink in 200 pairs, so the contest calls the direction correctly 71.3% of the time, a sign test at $p = 4 \times 10^{-13}$, and does better than chance in 18 of 21 categories. Roughly three pairs in ten move the other way, becoming *more* extreme under restriction, which no contest at any fixed noise scale can produce.

Third, the magnitudes tell two different stories depending on how they are aggregated, and the distinction matters. Inverting each pair for the noise scale that would reproduce its observed shift gives a median of 1.05 $[0.76, 1.31]$, so the typical pair is described by Case V about as well as one could ask. The mean implied scale is 2.49, with an interquartile range of 0.44 to 2.03, and it is that heavy upper tail, not the typical pair, which makes the mean shift four times larger than Case V predicts. Machine restriction behaviour is therefore not a uniformly exaggerated contest. It is a contest for most pairs and something considerably more violent for a substantial minority.

Fourth, and cautionary for the whole enterprise of asking machines about their preferences: listing order moves the same log odds by 1.27 nats, from -1.784 to -0.516, five times the effect the contest predicts and comparable to the effect under study. Both orders reject the axiom, so the finding is not an artefact, but any study that names alternatives in a fixed order measures position preference alongside choice. Order crossing is a precondition for the measurement meaning anything, not an optional check.

The distractor condition deepens the puzzle rather than resolving it. Adding one low-probability third option shifts the same odds by -4.017, further from the unrestricted value than the pair itself. Under either family the triple should lie *between* the pair and the full set, being intermediate in composition. It does not. Naming a small set appears to restructure preference rather than to prune it.

4 Machine preference is not order-stable

The odds result says the axiom is violated. A second measurement says something stronger, and it constrains every model in the paper rather than one of them.

Both families under comparison are rank-preserving. Renormalization divides by a constant, and contestant removal is a monotone transformation of the location vector, so each maps the unrestricted ordering of the survivors onto itself. Fitting a temperature, a noise ratio, unequal variances or heavier tails changes none of this: every member of either family, and of the wider random-utility class, predicts that if the model preferred i to j before the restriction it prefers i to j after it. A single reversal is outside the reach of all of them at once.

Reversals are not rare. Over the sweep's 37,764 cells, 32.8% of surviving pairs are discordant between the unrestricted and restricted distributions ($[31.6\%, 34.0\%]$ clustered by question type), 86.4% of cells contain at least one reversal, and in 48.0% of cells the leading survivor is not the one the unrestricted distribution ranked first. Restriction does not reweight a stable order. It reorders.

The natural objection is that reversals among near-tied, low-probability items are measurement noise. The data say the opposite: reversals concentrate among exactly the pairs where noise cannot explain them.

Pairs included	Pairs	Discordance
All	475,365	0.321 $[0.310, 0.333]$
Both items above 2% of restricted mass	87,402	0.390 $[0.380, 0.400]$
Both above 5%, unrestricted odds differ by $> 20\%$	48,885	0.409 $[0.400, 0.419]$
Both above 10%, unrestricted odds differ by $> 50\%$	24,932	**0.421** $[0.411, 0.431]$

Table 3: Rank discordance between the unrestricted ordering and the observed restricted distribution, clustered by question type. Restricting attention to high-mass, well-separated pairs raises the reversal rate rather than lowering it, which is the opposite of what a noise explanation requires.

Among pairs where both items carry at least a tenth of the restricted mass and whose unrestricted odds differed by more than half, 42.1% reverse. These are the comparisons a choice model should find easiest.

Three consequences run through the rest of the paper. The first is that no member of either family can be more than approximately right, so the remaining question is which is less wrong and why, which is how Section 8.6 onward should be read. The second is that a rule which declines to use the ordering is not obviously foolish, and indeed Section 8.7 finds that spreading mass uniformly over the survivors beats both structural rules; with a third of the ordering inverted, using it confidently is a liability. The third is that the optimal fitted flattening of $\gamma \approx 0.22$ acquires an interpretation: it is roughly what one should do with an ordering that is unreliable about a third of the time. The scale change that dominates the forecasting comparisons is the correct response to instability, not a separate phenomenon.

Section 8.10 reaches the same conclusion from a different direction. Asked for a least favourite rather than a favourite, models name items they never mention when asked for a favourite, so the two directions barely share a candidate set and neither family predicts one from the other. There is no single stable scale to read in two directions, just as there is no stable order to reweight.

5 Experimental design: masked language models

We probe the token probabilities of BERT [19] and RoBERTa [21], prompted to fill in blanks that invite choices among items of a category. Two designs are used.

In the first, the model answers an *original* question and a *qualified* question whose only difference is an adjective that shrinks the choice set:

1. *Original:* “My favorite state in the U.S. is [MASK] and I try to visit once a year.”

2. *Qualified:* “My favorite Western state in the U.S. is [MASK] and I try to visit once a year.”

[1]

[1] No qualifier partitions a category cleanly. Some Western states are more western than others, Oklahoma and Texas being the obvious hard cases, and a reader who disagrees with any particular assignment is probably right. The same fuzziness attends every design here: senses of a polysemous word overlap, “a random X” is not a uniform draw, a two-slot continuation invites contrast and collocation as well as exclusion, and an adjective alters

In the second, we eliminate a single option explicitly, feeding the top answer of a first prompt back into a second:

1. *First:* "My favorite primary color is [MASK] because it is SOMETHING."
2. *Second:* "My two favorite primary colors are [ANSWER] and [MASK] because they are SOMETHING."

where SOMETHING ranges over many adjectives so that luck of phrasing is averaged out. The ninety-nine categories and their adjective lists are committed as `research/polysemy_pilot/qvols/*.yaml` in the `winning` repository, and the sweep that consumes them is `research/polysemy_pilot/sweep.py`.

Both models make zero-parameter predictions of the restricted distribution from the unrestricted one. Luce's prediction is renormalization, Equation (1); in the two-step design this is the simplest Plackett–Luce [23] or Harville [20] construction. The Thurstonian prediction calibrates locations $\{a_i\}$ to the unrestricted token probabilities via $\mathrm{Prob}(X_i < X_k \; \forall k \neq i) = p_i$, removes the excluded contestant, and recomputes winning probabilities.

6 Results for masked language models

Table 4 shows the "favorite Western state" prompt for BERT. The Thurstonian column tracks the actual restricted probabilities more closely than Luce renormalization, which overweights the favorite (California) at the expense of the field. Table 5 repeats the exercise for "favorite girl's baby name", where the RMSE using Luce's probabilities is roughly 50% higher.

State	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
California	10.89	29.87	25.38	20.17
Arizona	6.74	18.50	17.20	10.00
Texas	5.50	15.10	14.59	7.68
Colorado	3.18	8.74	9.38	7.81
Oregon	2.89	7.92	8.67	8.57
Oklahoma	1.97	5.41	6.37	6.64
Nevada	1.66	4.55	5.54	10.18
Montana	1.59	4.37	5.36	14.27
Idaho	1.05	2.87	3.82	4.99
Wyoming	0.98	2.69	3.62	9.69

Table 4: Predicted probabilities under Luce's Choice Axiom and the Thurstonian model for the "favorite Western state" prompt (BertForMaskedLM). Both predictions use only the unqualified token probabilities; the Thurstonian model is closer to the actual qualified probabilities.

what the question means as well as which answers remain admissible. So none of these manipulations is a surgical deletion from a fixed choice problem, and machine restriction is not a pure urn problem in any case. What the designs require is weaker: that the manipulation removes an item from serious contention, and that both candidate laws face the identical stimulus. A qualifier whose boundary is arguable adds noise to the comparison rather than favouring either family, and the batteries are large enough, and the restriction phrasings varied enough, that no single fuzzy boundary carries the result. The exclusion phrasing of Section 8.6 is the cleanest instrument, and it is reported separately throughout.

Name	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
Lily	1.56	20.35	16.18	14.52
Emily	1.15	15.06	13.31	16.29
Mia	0.86	11.25	11.02	13.18
Bella	0.85	11.12	10.93	8.84
Sarah	0.68	8.89	9.44	9.41
Rachel	0.59	7.66	8.56	7.15
Kate	0.59	7.63	8.54	8.44
Lauren	0.47	6.11	7.39	7.23
Brittany	0.46	6.04	7.33	7.55
Chloe	0.45	5.89	7.21	7.39

Table 5: The same comparison for "favorite girl's baby name". The RMSE using Luce's probabilities is approximately 50% higher than the Thurstonian RMSE.

Across the full panel of qualified-question and single-item-elimination experiments, the Thurstonian model provides the more accurate prediction in approximately 80% of cases, with its advantage concentrated where the qualification is substantial and the entropy of the unrestricted distribution is low.

7 A human benchmark

The dispute this paper joins was never about machines. It was about people, and it stalled because the quantity in question is hard to observe in people: a choice distribution must be estimated from repeated trials, and on any single menu the two families differ by less than the resulting standard errors. That is why the argument was settled by tractability rather than by measurement, and it is the reason a machine is worth asking. But a machine result is only worth having if the human quantity can be put beside it, and for the odds statistic it can, from data already collected.

Complete rankings supply both halves at once. A respondent's choice from any subset is the highest-ranked available alternative, so the population's distribution over the full set and over any pair are both computable without asking anyone a second question. We take every complete strict-order collection in PrefLib with at least fifty respondents and at most twelve alternatives: Kamishima's sushi rankings, Netflix ratings, and two perceptual discrimination sets, ten files and 90 pairs in all.

People violate the axiom. The mean shift is -0.191 $[-0.322, -0.069]$, $|\delta|$ exceeds 0.1 nats in 84% of pairs, and the sign is negative in eight of the ten files: restricting a choice to two alternatives moves their odds toward parity, as a contest predicts and as an urn forbids. The sushi collection alone gives -0.285 over 45 pairs from 5,000 respondents, and a separate parse of Kamishima's original distribution of the same data gives -0.317, which is the reproducibility one would hope for from two independent paths through differently packaged files.

7.1 Controlling for how extreme the odds were to begin with

The raw machine and human means, -1.150 and -0.191, differ by a factor of six, and most of that factor is an artifact. A pair whose odds start near parity has little room to move; a pair at ten to one has a great deal. Machines begin from more extreme odds than people, with a median absolute log odds of 1.50 against 0.87, so a common proportional discount would produce a larger raw shift in machines with no difference in behaviour at all.

The scale-free quantity is that discount. Write

$$\delta_{ij} = -\lambda \log \frac{p_i}{p_j}, \tag{5}$$

so that $\lambda = 0$ is the Choice Axiom, which preserves odds exactly, and $\lambda = 1$ is total abandonment of the prior ranking, leaving the pair at parity regardless of what the full set implied. Fitted through the origin, with intervals from a pair bootstrap:

	Pairs	Median |log odds|	Discount λ
Choice Axiom (Luce)			0 exactly
Case V predicts, people	90	0.87	0.298 $[0.255, 0.335]$
Observed, people	90	0.87	**0.389** $[0.308, 0.466]$
Case V predicts, machines	276	1.50	0.120 $[0.095, 0.147]$
Observed, machines	276	1.50	**0.690** $[0.585, 0.792]$
Prior ranking discarded			1

Table 6: The discount applied to prior odds when a choice is named down to a pair, with each population's Case V prediction computed from its own unrestricted distribution. For people the contest is approximately right and the axiom is not. For machines both are wrong, the contest by a factor of nearly six in the same direction.

Controlled this way the machine exaggeration over people is 1.77 rather than six, and it survives: the intervals are disjoint. Within bands of starting odds the ordering is the same at every level, with mean shifts of −0.04 against −0.02 below half a nat, −0.26 against −0.07 from half to one, −1.07 against −0.28 from one to two, and −2.36 against −0.65 above two.

7.2 The century-old question, answered twice and differently

Because λ is zero under the axiom and positive under a contest, it scores both families at once, and each population can be compared against a contest calibrated to its own unrestricted distribution.

Population	Construct	Pairs	λ	Ratio to Case V
People	perceptual discrimination	48	0.188	1.16 $[0.85, 1.44]$
People	humour, 24,953 raters	45	0.393	1.12 $[0.90, 1.35]$
People	preference (sushi, Netflix)	60	0.420	1.33 $[1.10, 1.54]$
Machines	open-vocabulary recall	276	0.690	**5.77** $[4.32, 7.65]$

Table 7: The discount on prior odds, and its ratio to the discount a Case V contest predicts for the same distribution. That ratio has a null of exactly one: a simulated Gaussian contest returns 1.00 at every noise scale from 0.5 to 3.0, because calibrating locations to the observed distribution absorbs the scale. Two of the three human constructs are indistinguishable from one. Machines are not.

For people the classical dispute has a clear answer on this statistic, and the answer is Thurstone's. The ratio of the observed discount to the contest's own prediction is 1.16 $[0.85, 1.44]$ on perceptual discrimination, 1.12 $[0.90, 1.35]$ on humour and 1.33 $[1.10, 1.54]$ on food and film preference. Two of the three include one; the third barely excludes it. Human choice under restriction is, within the precision available, a Gaussian contest, and it is emphatically not renormalization, whose prediction of $\lambda = 0$ lies many standard errors away in every construct.

That ratio is worth dwelling on because it has an exact null. If a population really is a Gaussian contest at any noise scale, the ratio is one: simulating populations at scales from 0.5 to 3.0 returns 1.00 each time, since calibrating locations to the observed distribution absorbs the scale entirely. The statistic therefore cannot be inflated by a population being noisier or sharper than Case V, only by its being a different kind of process. Recovering 1.00 on simulated contests also serves as a positive control on the whole pipeline.

Machines return 5.77 $[4.32, 7.65]$. They are not a noisier contest or a sharper one; they are not a contest. Together with the rank reversals of Section 4, which no random utility model of any kind permits, the verdict is that the century-old dispute is settled one way for people and does not apply to machines.

A limit on precision is worth stating because it points at what would sharpen this. The intervals above are wide not because respondents are few, Jester contributes almost twenty-five thousand, but because a dataset of K alternatives yields only $K(K-1)/2$ pairs, so 45 to 60 pairs per construct is all that ten or twelve items can give. Precision here scales with the number of domains, not with the number of people in any one of them.

The humour data deserve a note on why they are trustworthy. Jester asks every user to rate the same ten jokes on a continuous scale before showing them anything else, so 24,953 complete rankings arrive with nothing imputed, and because the scale is continuous the tie convention that destroyed the election estimate barely applies: splitting equal ratings gives 0.393 and dropping them gives 0.390.

For machines it has no answer, because neither family is close. The contest predicts 0.120 against an observed 0.690, an overshoot of 5.8 with intervals nowhere near each other. Machines depart from the axiom in the direction the contest specifies and then continue well past it, which together with the rank reversals of Section 4 places their behaviour outside the random utility class rather than at some point within it. Note also that the human and machine overshoots differ in kind: people exceed a prediction of about 0.3, machines exceed a prediction of about 0.12, and the contest predicts less for machines precisely because their distributions are more concentrated, so their calibrated locations lie further apart and a two-horse field compresses them less.

7.3 Why real elections cannot be used here

Ranked-ballot elections would be the natural place to look for human choice at scale, and PrefLib holds over fifty such files with more than a million and a half ballots between them. They cannot answer this question, and the reason is worth recording because it is not obvious.

Voters rank a few candidates and leave the rest unranked, which the file format represents as a final tied group. That forces a choice with no defensible answer. Treating the tie as indifference, so that a restricted pair drawn from a voter's unranked candidates splits evenly, drives restricted odds toward parity and gives $\lambda = 0.680$ $[0.661, 0.699]$. Dropping tied comparisons and counting only pairs a voter actually ordered gives $\lambda = 0.147$ $[0.120, 0.175]$. The statistic swings by a factor of nearly five on a bookkeeping decision, because the first reading manufactures indifference and the second selects for engaged voters.

Complete rankings avoid the problem entirely: every respondent orders every alternative, so no truncation model is required and nothing is imputed. The human figures above therefore rest on complete-ranking data only, and elections are excluded rather than pooled. Using them would need a model of ballot truncation, which is a separate paper.

8 Which wrong rule is least wrong

Sections 3 and 4 rule out both families as literal accounts: the axiom fails its own test, and a third of the ordering inverts, which no rank-preserving rule can express. What remains is a forecasting question with a practical answer. Given the unrestricted distribution and nothing else, which rule predicts the restricted one least badly, and does the answer depend on how the question is asked? The batteries below are large because the differences are not, and because a claim about which approximation to prefer should hold across question types rather than on average over them.

Cekinmez, Wu, Marjieh and Griffiths [25] recently turned an everyday property of language into a measurable distribution over meanings: give a model a bare polysemous word ("bolt") with no disambiguating context, sample many outputs, and classify which sense each expresses. Across 32 text and image generation models they report a striking ladder of normalized sense entropies: image generation 0.10, text generation 0.25, human subjects 0.47, and, when models are asked to *predict* the distribution of senses, roughly 0.73. Models know the ambiguity; they do not express it. The finding sits within a growing literature on generative homogenization [26] and its sharpening under preference tuning [27]; their own ablation shows DPO alone cutting sense entropy from 0.25 to 0.18.

Read through the lens of Section 2, the ladder is what a contest model produces. Let the stated distribution proxy the latent locations $\{a_i\}$; let each generation be a contest among noisy sense performances. When noise is small relative to the spread of locations, winning probability is a steeply nonlinear function of location differences, sigmoidal rather than convex, and a mildly uneven prior collapses onto its mode. One noise parameter then indexes the entire ladder, and the "multimodal gap" becomes a noise gap rather than a knowledge gap. The rival explanation is a power tilt: renormalized p_i^γ, the Luce family with a temperature. Low-temperature decoding is literally this transformation of a fixed token distribution. Preference tuning and classifier-free guidance are often described in the same terms but are not the same operation: tuning changes parameters, and therefore logits, in a prompt- and token-dependent way, while guidance combines conditional and unconditional logits. The tilt is a rival account of the observed sharpening, not a description of what those procedures compute. The two families are distinguished not by the collapse itself but by behavior under choice-set restriction, precisely the instrument of Section 5, and an experiment run by neither paper until now.

8.1 Replication on a chat model

The smallest of our designs, and the only one using sampling rather than exact probabilities, replicates their generative setup on Claude models (October 2025 generation): 15 polysemous words from the stimulus set of [25], 150 independent sentence generations per word, senses classified by a second model against the closed inventory, plus a stated distribution elicited with their "predicting itself" framing. The collapse replicates almost exactly: mean normalized sense entropy 0.252 generated (their text-model figure: 0.25) against 0.880 stated.

Fitting the two one-parameter families from stated to generated distributions gives a statistical tie (Thurstone $\sigma = 0.42$, RMSE 0.273; power tilt $\gamma = 2.25$, RMSE 0.273; Thurstone the better fit on 10 of 15 words), and the reason both fit poorly is itself informative: for several words the generated mode is not even the stated argmax ("port" generates *harbor* in every sample while rating *connector* most probable). Stated reports are a biased window on the latent locations, not merely a flattened one. This is consistent with [25], whose stated distributions overestimate even human diversity. Calibration should therefore use revealed behavior, which is what the restriction design does.

8.2 Choice-set restriction

For the six words whose unrestricted distributions were non-degenerate, we re-sampled with the modal sense excluded by instruction (150 samples per condition) and compared zero-parameter predictions of the restricted distribution: Luce renormalization against Thurstonian contestant removal, both calibrated to the add-half smoothed unrestricted sense counts.

Word	Luce RMSE	Thurstonian RMSE	Restricted distribution
bolt	0.450	**0.413**	fastener .68, run .18, lock .14
seal	**0.077**	0.089	animal .67, close .33
pitch	0.655	**0.630**	throw .06, tone .31, field .63
bat	0.559	**0.546**	animal 1.00, hit .00
match	0.758	**0.651**	firestick .82, contest .18
iron	**0.002**	0.037	metal .95, press .04, golf .01
pooled	0.495	**0.459**	

Table 8: Restriction test on polysemous words (Claude Haiku 4.5, 150 samples per condition, modal sense excluded by instruction). The Thurstonian prediction is closer pooled and on four of six words (bootstrap over words: $P(\text{Thurstone better}) = 0.90$).

The margin favors Thurstone, but the qualitative signature matters more than the margin: under restriction the model does not spread probability proportionally, as Luce requires. It *re-collapses* onto a new modal sense (match $\rightarrow$ *firestick* at .82; iron $\rightarrow$ *metal* at .95; bat $\rightarrow$ *animal* at 1.00). Remove the winning horse and the runner-up now wins nearly always. That is low-noise contest behavior, in the same low-entropy, heavy-qualification regime where the Thurstonian advantage concentrated in Section 6. Some rank flips under restriction exceed even the Thurstonian sharpening, and inventories with overlapping senses ("bat") contaminate the exclusion instruction; neither family survives those cells.

8.3 Exact probabilities on GPT models

Sampling noise can be removed entirely on models that expose token log probabilities. We ported the qualified-question design of Section 5 to GPT-4o-mini, GPT-4o and GPT-4.1: twenty category pairs ("my favorite color is" / "my favorite warm color is"), two phrasings, next-token distributions read exactly from the top-20 log probabilities, with the Thurstonian field restricted to a declared category inventory. Across 71 usable cells the comparison is a statistical tie with a Thurstonian lean: Thurstone wins 38 of 71 cells and has the lower pooled RMSE on all three models (e.g. 0.290 vs 0.295 on GPT-4o), but a bootstrap over cells does not separate the families ($p = 0.29$).

The instrument, however, has weakened for an interesting reason: the unqualified distributions of preference-tuned chat models are already near-degenerate ("my favorite tree is" puts 0.94 on *oak*), so most cells carry little discriminating information. Under either model the surviving favorite keeps nearly everything, and the comparison hinges on tail probabilities. The masked models of Section 6 answered the unqualified question with entropy to spare (California at 10.9%); their preference-tuned descendants have already collapsed before the experimenter arrives. This is the same collapse measured by [25], now visible as a loss of experimental power: alignment has pushed the interesting variation into the tails, where only exact log-probability designs can reach it.

8.4 Random elicitation restores the regime

The remedy is to ask for randomness rather than preference. "Name a random programming language" followed by explicit single-item elimination raises mean unqualified entropy to 0.38 and returns the experiment to the regime where the two families disagree. The elimination takes one of two forms, either an exclusion ("... that is not Python") or the two-slot continuation of Section 5 ("I already picked one random programming language: Python. Name another..."). The recovery is only partial, since models are poor samplers even on request, with "a random planet" putting essentially all mass on *Mars*. The homogenization of [26] extends to explicit randomness.

To control for phrasing and for which item is removed, the definitive battery deletes *each* of the top eight observed items in turn, under two phrasings, across 50 categories and three GPT models. Thirty-one categories retain enough inventory-matched mass to be scorable, giving 692 permutation-controlled cells on the three headline models and 1,195 once the two cheaper models are added, scored by the proper rule KL(actual $\|$ prediction), with bootstrap confidence intervals over cells.

Stratum	n	mean ΔKL (Luce − Thurstone)	95% CI
all cells	692	+0.025	[+0.016, +0.035]
non-degenerate prior ($\tilde{H} > 0.2$)	521	+0.036	[+0.026, +0.048]
degenerate prior ($\tilde{H} \leq 0.2$)	171	−0.008	[−0.025, +0.011]

Table 9: Permutation-controlled deletion battery, exact log probabilities, KL scoring. Positive favors Thurstone. The advantage is present overall and concentrates where the model retains genuine choice entropy.

The contest is the better description wherever machine preference is genuinely stochastic, and renormalization is competitive only in the degenerate cells that preference tuning has emptied of choice.

8.5 The original stimuli at scale

The largest battery, and the largest effect, comes from re-running the *original* stimulus set of this paper's 2024 version. That is 97 scorable categories crossed with every adjective in the original appendix, 852 distinct prompt pairs, each put to five GPT models in the original two-slot form ("My two favourite human organs are the [ANSWER] and the [MASK] because they are SOMETHING") with exact log probabilities. Across the 2,183 cells of the three headline models the conditional second choice favors the contest model by the largest margin anywhere in the paper: mean ΔKL +0.55, 95% CI $[+0.51, +0.60]$, and in both entropy strata. Adding the two cheaper models brings the battery to 3,660 cells and +0.52 $[+0.49, +0.55]$. Predicting the runner-up given the winner is precisely where Luce renormalization fails hardest. It is the machine-preference analog of the Harville [20] overpricing of exactas that motivated the Thurstonian computation in the first place. The same questions asked of BERT in 2024 and of GPT-4-class models in 2026 return the same verdict, with a larger margin under exact measurement.

8.6 Every question type, not merely every model

A verdict that holds on average across categories is weaker than one that holds within each of them, so the broadest battery enumerates the design space rather than sampling it. All 99 question types of the 2024 appendix, crossed with all 926 of their adjective variants, under two restriction designs, deleting each of the top eight observed items in turn, on the three cheaper models: 37,764 scored cells from 45,848 logged measurements. Item fields are open-vocabulary, taken from whatever the model itself offers at the unqualified prompt, so no hand-authored inventory can select the categories that suit either family.

Thurstone is ahead in 29,307 of 37,764 cells, a rate of 77.6% that sits almost exactly on the 80% reported for BERT in the 2024 version, now with two orders of magnitude more cells. Mean ΔKL is +0.1495 $[+0.1451, +0.1539]$. Both restriction designs favor the contest model, the two-slot continuation at +0.199 more strongly than the exclusion phrasing at +0.109, and so does every model: +0.111, +0.140 and +0.198 for GPT-4.1-nano, GPT-4o-mini and GPT-4.1-mini.

The breakdown is more informative than the pooled figure. **All 99 question types favor Thurstone** on average, not 90 or 95 of them. There is no category of question, among organs, colors, cities, sports, instruments and the rest, in which renormalization is the better account. The deficit also concentrates where the theory says it must: deleting the model's favorite costs Luce +0.407, against +0.114 for deleting a minor item. Removing the winner is the case Harville [20] mispriced, and it is where the urn and the contest diverge.

At this sample size the degenerate stratum also separates from zero (+0.142 against +0.157 for non-degenerate cells), which sharpens the earlier reading. Preference tuning does not repeal the contest structure in collapsed distributions; it only leaves too little entropy for a smaller battery to detect it.

Cells nest inside question types, adjective variants, base prompts and models, so cell-level bootstrap intervals overstate the independent information. A cluster bootstrap that resamples the 99 question types and retains every cell belonging to a selected type gives $[+0.135, +0.165]$ against the cell-level $[+0.145, +0.154]$. Clustered intervals are the ones reported in Table 1 for every design; here the clustered version is barely wider, because the effect is present within essentially every cluster rather than concentrated in a few.

8.7 What the comparison does and does not identify

The convention behind every comparison so far bears restating. The contest is independent performances of equal unit variance, which is Thurstone's Case V, the default member of the family and the one Mosteller [2] made practical. It is also the most constrained member. The family contains freer relatives, among them unequal variances, correlated performances and heavy-tailed noise, and any of them would fit better by construction.

Searching that space and reporting the winner would be unfair to Luce, which in its bare form has nothing to fit. So nothing is tuned. Every prediction in every battery is computed from the unrestricted distribution alone, with no free parameter and no training set, which is what makes the cells out-of-sample in the strong sense: neither family ever sees a restricted distribution before predicting it. The price of that discipline is that a positive result cannot be attributed to Gaussian geometry specifically. It shows that the default contest forecasts better than the default urn.

Nor does winning that comparison establish that a Gaussian race generates the data. Two further baselines settle what the comparison is worth, and one of them is unflattering.

The first is trivial and parameter-free: ignore the unrestricted distribution altogether and spread mass evenly over the survivors, $q_i = 1/|S|$. This is the $\gamma \to 0$ limit of the tilt family, it needs no training data, and it beats both structural rules by a wide margin. Over the sweep's 37,764 cells its mean KL is 1.089 against 1.749 for Case V removal and 1.898 for renormalization, and it is ahead in 94 of the 99 question types. A rule that discards the model's own ranking predicts the restricted distribution better than either rule that uses it.

The second grants each structural family one parameter. For the Luce side that is the tilt of Equation (**??**). For the contest side the parameter has to be introduced carefully, because a Thurstone model whose locations are recalibrated at a new scale is scale-invariant: if b_i are the unit-variance locations then $a_i = \sigma b_i$ reproduces the same unrestricted probabilities and the same deletion predictions. The parameter is therefore not a variance but a *restriction-stage noise ratio*. Locations are calibrated from the unrestricted distribution at unit noise,

$$p_i = \Pr\big(a_i + Z_i = \min_j (a_j + Z_j)\big), \qquad Z_j \sim N(0,1), \tag{6}$$

those locations are then held fixed, and the restricted stage is predicted with the noise inflated by σ:

$$T_i^\sigma(p, S) = \Pr\big(a_i + \sigma Z_i = \min_{j \in S} (a_j + \sigma Z_j)\big). \tag{7}$$

So σ measures how much noisier the contest becomes when the choice set is named, exactly as $1/\gamma$ measures how much flatter the urn becomes.

Forecast	Information used	Held-out mean KL
Renormalization	unrestricted distribution	1.926
Case V contestant removal	unrestricted distribution	1.786
Uniform over survivors	survivor set only	1.089
Tilted renormalization, $\gamma \approx 0.22$	plus one fitted scalar	0.803
Contest at noise ratio $\sigma \approx 3.4$	plus one fitted scalar	**0.794**

Table 10: Five forecasts of the restricted distribution on held-out question types. Both parameter-free structural rules are beaten by a uniform rule that uses no ranking information at all, because both are far too concentrated. Granting either family a single scale parameter recovers the ranking information and overtakes uniform by a wide margin. Fitted values come from grouped five-fold cross-fitting over question types, refitted in every fold.

One further decomposition settles how much of the contest's advantage is geometry and how much is scale. Both renormalization and contestant removal are rank-preserving monotone maps of p, so each can be tilted to its own best exponent, and if the contest's edge were purely that it injects entropy the two tilted forecasts would coincide. They very nearly do: tilted renormalization reaches 0.8028 and tilted contestant removal 0.8046, a difference of -0.0017 $[-0.0044, +0.0009]$ that does not exclude zero.

Isolating shape from scale directly gives the same answer with the sign restored. Rescaling every prediction cell by cell until its entropy equals the entropy of the observed restricted distribution removes the scale question entirely, leaving only where each rule puts the mass. Under that oracle rescaling the contest is ahead by $+0.0047$ $[+0.0026, +0.0071]$, favouring it in 63 of 99 question types. The geometry is real and it is small.

Comparison	Luce side	Thurstone side
No parameters	1.926	**1.786**
No parameters, no ranking used (uniform)	1.089	
One fitted scale, cross-fitted by type	0.803	**0.794**
Oracle per-cell entropy match, shape only	2.341	**2.337**

Table 11: Decomposing the advantage. Held-out mean KL, lower better. The parameter-free gap of 0.140 shrinks to 0.009 when each family may set one scale, and to 0.005 when scale is removed altogether by matching entropy cell by cell. The scale change is the phenomenon; the Gaussian geometry is a small consistent residual. Entropy matching is not a KL oracle, which is why its absolute values are worse: forcing a prediction to the observed entropy is costly wherever the two disagree about which survivor wins.

How small is small? The natural yardstick is the setting that motivated the Thurstonian machinery in the first place. On 5,079 paired horse races, with both models inverted from the identical market win vector so that any difference at second place and beyond is purely the ranking model, the Thurstonian order statistics beat Harville's renormalization by $+0.0018$ in log loss at position two, $+0.0139$ at position three and $+0.0222$ at position four.[2] Those are the differences that make Thurstonian pricing worth the computation in a real market.

Our geometry residual sits inside that range and can be placed within it. At matched entropy the $+0.0047$ is 2.6 times the place-two difference, 34% of place three and 21% of place four; at matched parameter count the $+0.0085$ is 4.9 times place two, 64% of place three and 40% of place four. Its statistical strength is comparable to place three, an interval excluding zero at roughly four standard errors against racing's 3.9.

Measured instead against the predictive error that remains, the ordering reverses, because the fitted machine forecast is far more accurate in absolute terms than a place probability: $+0.0085$ on a base of 0.794 nats is 1.07% of remaining loss, against 0.91% for the place-four difference on a base of 2.436. Absolute nats is the more conservative yardstick and the relevant one for a wagering edge, since log wealth is linear in log loss, so the fair summary is a third of the place-three effect. On either yardstick the Gaussian contribution in language models is small beside the scale effect that dominates the same data, and of the same order as the effect that has sustained a literature and a betting edge. Both statements are true and the second is easily lost.

The contrast with depth is instructive in the other direction. In racing the advantage grows as places are filled, because the field stays large; in open-vocabulary elicitation it shrinks with

[2] Measured with the companion code to [18]; log loss per finish position, paired by race. The comparison with our figures is one of magnitude rather than of identical construction, since a place probability and a restricted choice distribution are different objects, but both are scored in nats on identical inputs.

depth, because each removal shortens the field until only two or three candidates remain. The mechanism is the same and the experimental economics are opposite.

Read together, Tables 10 and 11 say something more specific than the headline comparison. What prompted restriction does to a language model is mostly not a reallocation among survivors in proportion to anything. It is a large increase in entropy: an effective noise inflation of about 4.5 on the Gumbel scale, or 3.4 on the Gaussian one. Renormalization cannot express that at all, since it holds odds fixed by construction, which is why it loses everywhere. Contestant removal expresses part of it, because removing a contestant and re-running the race necessarily spreads mass, and that is the bulk of the +0.150 advantage the batteries measure. On top of the scale effect sits a small, statistically detectable and cross-category consistent contribution from the Gaussian shape itself.

The recommendation follows the decomposition rather than the headline. With no historical restriction data, contestant removal is the better of the two classical defaults, in every design here and in all 99 question types, and renormalization should be abandoned; a practitioner should also know that a uniform prior over survivors beats both, and that the reason is scale. With any historical restriction data at all, even one scalar estimated once, a scaled forecast dominates every unfitted rule by about a nat, and the scaled contest edges the tilted urn.

What none of this establishes is that a stable Gaussian race runs inside the model. Several untested alternatives would produce similar reallocation, among them heteroscedastic or correlated races, heavy-tailed contests, similarity-sensitive nesting, and retrieval dynamics specific to list completion. The identified claims are about forecasting and about scale, not about mechanism.

8.8 The deficit grows with model tier

Across the five models the margin is ordered by tier: GPT-4.1-nano +0.30, GPT-4o-mini +0.35, GPT-4o +0.42, GPT-4.1-mini +0.65, GPT-4.1 +0.88. The more capable the model, the further its revealed preferences sit from the Choice Axiom its own output layer suggests.

The architecture is constant along that ladder, which narrows the candidate explanations without settling them. All five models emit a softmax, and a softmax implements Equation (1) mechanically: mask a logit, renormalize, and the Choice Axiom holds by construction. The deviation therefore cannot originate in the output layer. It originates in what the network computes before the output layer, and it grows with capability, as though the hidden layers were working against their own parameterization and getting better at it.

The obvious confound is concentration. More capable models answer the unqualified question more sharply (mean $\tilde{H}$ falls from 0.276 on GPT-4.1-nano to 0.105 on GPT-4.1 over matched prompts), and sharpness alone widens the gap between an urn and a contest, since the contest map amplifies rank separation. Three controls bear on the two explanations, all run on the 512 prompts measured on every model, paired within the identical question. None of them holds concentration fixed continuously, so the stratification below is a coarse control rather than a matched comparison.

Holding the question fixed, holding concentration fixed, and holding the size class fixed while advancing one model generation all leave the effect intact. Sharper locations are therefore part of the story but not the whole of it: at matched concentration the frontier model still departs further from renormalization than the cheapest one.

The panel admits one further decomposition, into a generation axis and a scale axis, and the two do not behave alike. Advancing a generation at fixed size class moves the deficit reliably, by $+0.478$ $[+0.348, +0.613]$ for the flagships and $+0.276$ $[+0.172, +0.380]$ for the mini class. Varying

Paired comparison (same question prompt)	n	ΔKL difference
GPT-4.1 − GPT-4.1-nano	512	$+0.571$ $[+0.426, +0.721]$
GPT-4.1 − GPT-4o	512	$+0.478$ $[+0.348, +0.613]$
GPT-4.1-mini − GPT-4o-mini (same size class)	512	$+0.276$ $[+0.172, +0.380]$
GPT-4.1 − GPT-4.1-nano, both degenerate	142	$+0.717$ $[+0.387, +1.067]$
GPT-4.1 − GPT-4.1-nano, both non-degenerate	144	$+0.363$ $[+0.191, +0.548]$

Table 12: The tier effect survives matching on the question and, coarsely, on concentration. Positive means the more capable model departs further from Luce renormalization.

scale within a generation moves it too, by $+0.253$ and $+0.318$ for the two steps down the 4.1 family, but the corresponding step within the older family, GPT-4o against GPT-4o-mini, is $+0.050$ $[-0.037, +0.140]$ and does not exclude zero. Within that generation, in other words, making the model larger left its departure from the Choice Axiom essentially unchanged, while moving the same size class one generation forward changed it substantially. Whatever separates the two generations appears both to raise the deficit and to make it sensitive to scale.

We report this as a pattern in five models rather than a scaling law, and note two limits. Size class here is a vendor label, not a parameter count, and tier is a proxy for capability rather than a measurement of it. The pattern does, however, yield a falsifiable prediction: if what separates the generations is post-training rather than scale, a base model should obey the Choice Axiom more closely than its instruction-tuned sibling. Section 8.9 tests that prediction directly, on models whose base and instruction-tuned checkpoints are both available.

8.9 Preference tuning moves the deficit, in some families

The commercial models expose no base checkpoints, so this test uses open-weight pairs run locally, where the whole vocabulary is available and nothing is truncated. Base and instruction-tuned siblings receive the identical completion prompt, with no chat template on either side, so the checkpoint is the only difference. Both are given the stimuli of the sweep in Section 8.6 under its two-slot design, so that nothing about the measurement varies either.

Contrast (paired within question, adjective and deleted item)	n	ΔKL step	95% CI
Instruction tuning, Qwen2.5 0.5B	751	$+0.0026$	$[+0.0019, +0.0034]$
Instruction tuning, Qwen2.5 1.5B	627	$+0.0038$	$[+0.0017, +0.0060]$
Instruction tuning, Qwen2.5 7B	809	$+0.0247$	$[+0.0193, +0.0305]$
Instruction tuning, Mistral 7B	171	$+0.0194$	$[+0.0122, +0.0286]$
Instruction tuning, gemma-2 2B	453	-0.0003	$[-0.0008, +0.0001]$
Instruction tuning, Llama-3.1 8B	408	-0.0010	$[-0.0016, -0.0004]$
All six pairs pooled	3,219	$+0.0084$	$[+0.0069, +0.0100]$
Scale, base only, 0.5B to 1.5B	410	$+0.0025$	$[+0.0018, +0.0033]$
Scale, base only, 1.5B to 7B	484	$+0.0017$	$[-0.0000, +0.0033]$
Scale, base only, 0.5B to 7B	261	$+0.0039$	$[+0.0022, +0.0055]$

Table 13: Instruction tuning against the matching base checkpoint, and scale among base checkpoints. Positive favors Thurstone. The tuning effect is large in two families, absent in a third, and slightly reversed in a fourth.

The prediction holds in four of the six pairs and fails in two. Tuning moves Qwen2.5 away from Luce at all three sizes, by an amount that grows with scale from +0.0026 to +0.0247, and moves Mistral 7B by +0.0194. It does nothing detectable to gemma-2 2B, and it moves Llama-3.1 8B slightly the other way, by −0.0010 with an interval excluding zero. Pooling all six gives +0.0084 $[+0.0069, +0.0100]$, but that pooled figure rides on Qwen and Mistral and should not be read as a general law.

Where tuning does move a model, it moves it further than scale does. Among Qwen2.5 base checkpoints the entire span from 0.5B to 7B is +0.0039, about a sixth of what tuning contributes at 7B alone. So post-training is capable of dominating parameter count in setting how far a model sits from the Choice Axiom, which is the mechanism the commercial generation effect of Section 8.8 suggested. But the two null and reversed families show that it does not do so by necessity, and whatever distinguishes Qwen and Mistral post-training from Llama and gemma post-training is not visible from outside the labs.

Preference tuning is therefore one route by which a model departs from renormalization, substantially so in some families, and the direction of the effect is not fixed across families. That is consistent with the DPO ablation of [25], which changes sense entropy with no change in size, without implying that every tuning recipe pushes the same way.

Three further caveats bound all of these numbers. The absolute magnitudes are one to two orders of magnitude below the commercial panel on identical stimuli, so the local and commercial ladders are separate evidence rather than one curve. The weights are 4-bit quantized, which could compress the tails where the families disagree. And two of the six pairs sit so close to zero deficit overall that the paired contrast has little room to move in either direction.

One further caution emerged with practical import for all such experiments: the "actual" restricted distribution depends materially on how the restriction is phrased. Holding everything else fixed, the exclusion and two-slot variants disagree about the restricted distribution by a median maximum gap of 0.14, and up to 0.65 (GPT-4o-mini names *Earth* after *Mars* with probability 0.94 under one phrasing and 0.56 under the other). Under surgical removal from a fixed preference state the two phrasings would coincide; in a language model the choice-set manipulation is itself a prompt intervention. Framing effects of this size sit above both choice models, and any account of machine preference that ignores them is incomplete.

8.10 Ordered selection, depth, and reversibility

The two-slot design is an exacta. Asking instead for an ordered list, and eliciting each slot separately with the earlier picks fed back in, extends it to a trifecta and beyond, so that Plackett–Luce [23] renormalization and Thurstonian contestant removal can be compared at every depth. The contest calculation here is stagewise: the taken items are removed and the surviving field is re-raced, which is not the conditional order statistic of a single Thurstone draw. It suits successive prompts, where the model recomputes its distribution at every slot, but the racing vocabulary should not be read as the one-draw joint ranking model. Four framings were run over 40 categories and five models to depth five: two neutral ("select five random X in order", "name five different X, in order") and two preference-ordered (most favourite first, least favourite first).

The contest model wins under all four framings, which are different elicitation architectures rather than paraphrases: +0.55 most-favourite-first, +0.38 different-in-order, +0.26 random-order, +0.18 least-favourite-first, all intervals excluding zero over 1,363 cells.

Depth, however, does not behave as the racing analogy predicts. Since slot d involves $d-1$ removals and renormalization can only redistribute vacated mass proportionally, the Luce deficit might be expected to compound with depth. It does the opposite, falling from $+0.42$ at slot two to $+0.24$ at slot five, and within a given category and model the change from the second slot to the deepest is negative on average (-0.23, positive in only 161 of 395 ladders). The mechanism is visible in the cell sizes: each removal shortens the surviving field, and by slot five most cells have only two or three candidates left, where the two families cannot differ much. Depth in an open-vocabulary elicitation removes discriminating power rather than adding it, which is the opposite of the racing case, where the field stays large as places are filled.

Reversibility is a sharper test, and both families fail it. Under a Thurstonian scale, asking for the least favourite is the same latent vector read backwards, since the minimum of X is the maximum of $-X$; one calibration must then predict both directions with no free parameters. Luce's axiom, by contrast, says nothing about worst choice; replacing weights w_i by $1/w_i$ is an extra bipolar assumption rather than a consequence of IIA, and negating a Gumbel shock does not give another Gumbel shock. The two sides of this test are therefore not symmetric. Neither works: predicting the least-favourite-first distribution from the most-favourite-first one costs 5.3 to 9.0 nats depending on the model, for both families alike, and the ranking of models by this quantity does not track capability. The reason is that the two directions barely share an item set. Asked for a least favourite, models name items they never mention when asked for a favourite, so the reversed question is not the same contest re-read but a different one. Machine preference in this regime is not a single scale with two ends.

8.11 Menus, duplicates, and context effects

Presenting the choice set explicitly changes the phenomenon entirely, in three ways. The design is "pick one of these: ...", with menu order randomized and probabilities read from option-token log probabilities.

First, menu choice by preference-tuned models is nearly deterministic. A transport menu (car, bus, train, bike, walk, taxi) under contextual conditioning behaves with perfect semantic sense. "It is raining" sends GPT-4o to *taxi* with probability 1.00 and "there is a fuel shortage" sends it to *bike* at 0.99. But the resulting distributions are so degenerate that menu-based deletion cannot separate the choice models sharply. Across 229 deletion cells the contest is ahead by only $+0.026$ $[+0.008, +0.045]$, an order of magnitude below the open-vocabulary batteries and consistent with Table 9's stratification, since a distribution this concentrated leaves the two rules little room to differ. The determinism extends to simulated agents: fictional election candidates with platform descriptions, evaluated through fictional voter personas, produce near-certain votes even when the persona is written to be genuinely cross-pressured, and candidate-withdrawal cells are consequently uninformative. The model represents a persona as an argmax decider; it does not carry within-persona randomness at all.

Second, machines are anti-red-bus. In Debreu's classic objection to the Choice Axiom [28], splitting a bus into a red bus and a blue bus should not double the bus vote; humans treat near-duplicates as substitutes, which correlated-noise (probit) models capture. Across ten duplicate-pair families the models show little sign of substitution: the duplicated category's total share typically inflates, often beyond even the urn-doubling prediction (apple $0.08 \to 0.53$ when split into green and red), and a substitution baseline is the best predictor in only 15 of 41 cells against Luce's 24. (With exchangeable duplicates, Luce and independent Thurstone nearly coincide; the test's power is entirely against substitution.) Description salience, in that a *red* apple is more vivid than an apple, appears to dominate any shared-noise structure.

Third, where a menu does leave residual entropy, machine choice violates *regularity*, the property, shared by every random utility model including Thurstone's, that an added option cannot increase an incumbent's absolute share. Using two-attribute options (price versus quality) in randomized lettered menus, an asymmetrically dominated decoy [29] increased the target's absolute share in 16 of 30 family-model cells and a compromise manipulation [30] in 7 of 30, so 23 of the 60 tests violate regularity outright. The phone-plan family shows a textbook attraction effect (GPT-4o: target share 0.26 $\to$ 0.84 when its decoy appears). Machine preference, like human preference, is not a random utility model in the menu regime.

The menu regime admits a verdict stronger than any pairwise model comparison, because exact probabilities make it feasible to measure the *complete choice structure*: all eleven menus over a four-item universe, a measurement no human experiment delivers cleanly. By the Block–Marschak–Falmagne theorem [31, 32], such a structure is random-utility representable if and only if its Block–Marschak sums are nonnegative. Across six item families and five GPT models, **all 30 measured structures violate the conditions** (4–9 negative terms of 32 each; worst -0.77, far beyond measurement noise from order averaging). Fitting the structures anyway, the best possible random utility model (a distribution over all 24 rankings) leaves half the divergence unexplained (total KL 8.1, against 16.2 for the best Thurstone locations and 17.5 for the best Luce utilities, with Thurstone the better parametric fit in 26 of 30 structures). In the menu regime, machine choice is not a random utility model at all.

Together the batteries locate the choice law in the *elicitation architecture*. Open-vocabulary recall behaves like a Thurstonian contest among retrieved candidates (Table 9); explicit menus behave like near-deterministic maximization perturbed by salience and context effects that no random utility model accommodates. Between them sits the sampled-sentence regime of [25], where the contest reading again does best.

9 Discussion

The classical dispute has an answer here, and it is not the answer either side wanted. Luce's Choice Axiom is refuted on the one statistic where it makes a point prediction: naming a choice down to two items moves their log odds by -1.15 nats when the axiom requires zero, in both listing orders, and by more than a tenth of a nat in 96.8% of pairs. But the Thurstonian alternative is not thereby installed, because the failure is of a kind no random utility model can represent. A third of surviving pairs reverse rank, and the leading survivor changes in half of all cells, while every member of both families preserves order by construction. Restriction reorders machine preference; it does not reweight it.

What survives is a ranking among approximations, and it is consistent enough to be useful. The contest is less wrong than renormalization in all 99 question types of the largest battery, it calls the direction of the odds failure in 71% of pairs against an axiom that predicts no movement at all, and on the median pair its predicted magnitude is right. The reason is visible in the decomposition: renormalization holds odds fixed, a contest flattens by an amount that depends on the cell, and flattening is the correct response to an ordering that is unreliable. That is also why a uniform rule over survivors beats both, and why the best single fitted parameter is a heavy flattening rather than a mild one. Practitioners who mask an option and renormalize are using the worst of the rules examined here; those who want a parameter-free default should use contestant removal and expect it to be beaten by anything that learns the scale.

Two results owe nothing to the dispute and may outlast it. Listing order moves the same log odds by 1.27 nats, five times the effect either choice model predicts, so any study that names alternatives in a fixed order is measuring position preference alongside preference. And adding a single low-probability distractor to a named pair moves the odds -4.02, further from the unrestricted value than the pair alone, where both families require the triple to lie between pair and full set. We cannot account for that, and it is the clearest sign that naming a small set restructures the problem rather than pruning it.

The regime dependence is equally material. Everything above concerns open-vocabulary elicitation, where the model supplies its own alternatives. Present the alternatives explicitly and the phenomenon changes: choice becomes nearly deterministic, deletion barely separates the two families, near duplicates inflate rather than substitute, and all thirty measured choice structures violate the Block–Marschak conditions, which no random utility model of any kind can survive. There is no single law of machine choice to be found. There is a law-like regularity in one elicitation architecture and something closer to salience-driven maximization in another.

Conjectures, labelled as such

Three implications are worth stating and none is established here. Generative mode collapse may be the output of a low-noise contest over intact locations rather than a defect of knowledge, in which case diversity would be restored by inverting the contest and re-sampling at a chosen noise level rather than by raising temperature; the evidence is a fifteen-word exploratory design with six usable cells, and the intervention study that would test it, against temperature, top-p, typical and diversity-aware decoding, has not been run. A contest-structured output layer might generalize better than a softmax, and the gradient is not the obstacle, though the cost and normalization are.[3] And if the deficit grows with capability, as it does across five commercial models and with post-training in four of six open-weight pairs, then a Thurstonian diagnostic might serve as a label-free measure of model quality, except that the reversibility diagnostic does not order models by capability at all.

Limitations

The generative replication of Section 8 is the smallest design here and the only sampled one: fifteen words, one model, a single machine judge, temperature uncontrolled. The commercial batteries read a truncated candidate field, so their estimand is reallocation among previously serious candidates rather than over the whole vocabulary; the local batteries avoid this by scoring whole items over the full vocabulary, and their magnitudes cannot be pooled with the commercial ones. Local weights are 4-bit quantized. Model tier is a vendor label standing in for capability, and entropy stratification is a coarse control rather than continuous matching. The distractor anomaly and the reversibility failure are unexplained by anything in the paper.

None of these touches the two central measurements. The axiom fails its own test, and machine preference under restriction is not order-stable.

[3] The leave-one-out products that $P(X_i \text{ minimal}) = \int \phi(x-a_i)\prod_{j\neq i}(1-\Phi(x-a_j))\,dx$ requires are what the lattice method of [18] exists to avoid recomputing: prefix and suffix survival products over a common grid give every contestant's product in two passes, and the same arrays supply the derivatives, so the backward pass costs the same order as the forward one and no graph over the $n(n-1)$ survival terms is built. The obstacles are the forward cost, $O(nG)$ against a softmax's $O(n)$ with G a few hundred cells, the $O(nG)$ memory of the prefix and suffix arrays unless streamed, and the fact that a softmax-selected shortlist does not define a normalized full-vocabulary likelihood, so a target-plus-shortlist normalization would have to be worked out for training. We have not built it.

Data and code availability

Every battery, its stimuli, and every model response are in `research/polysemy_pilot` of `github.com/microprediction/winning`, which also contains the Thurstone lattice implementation used throughout. Each paid API response is written to an append-only JSONL log before anything is computed from it, so all derived numbers are recomputable offline: the logs total 61,512 measurements (`sweep_raw.jsonl`, `vol_raw.jsonl`, `perm_raw.jsonl`, `random_raw.jsonl`, `exact_raw.jsonl`, `bm_raw.jsonl`, `local_qvols_raw.jsonl` and others), and re-running any battery script re-scores from the logs without further inference. The item inventories are in `inventory.py`, the model tiers in `models.py`, and the tilt and clustered-bootstrap analyses in `power_tilt.py` and `one_param.py`.

Settings common to the commercial batteries: chat completions with `max_tokens=1`, `logprobs=True`, `top_logprobs=20`, `temperature=1.0`, no penalties, no seed. Models are the aliases `gpt-4o`, `gpt-4o-mini`, `gpt-4.1`, `gpt-4.1-mini` and `gpt-4.1-nano`, queried in August 2026; the logs record each response verbatim, but the aliases are moving targets and a future re-run need not reproduce them exactly. The GPT-5 tier cannot be used at all, since it refuses log-probability requests. Local models are 4-bit `mlx-community` conversions run with full-vocabulary logits.

Candidate handling, which matters for the comparison: item fields are the top-ten surviving tokens after lower-casing, stripping, and discarding non-alphabetic and stop-word tokens; probabilities of surface variants that normalize to the same word are summed; the retained mass is renormalized over the field, so an item's probability is its first-token mass rather than a full-sequence probability. Truncation is not neutral between the two families. Under renormalization, conditioning the winner to lie in a retained set equals deleting the omitted alternatives, whereas under a contest the two differ, so a Thurstone calibration fitted to renormalized top-k probabilities need not recover the locations of a coherent full-vocabulary contest. The local batteries avoid this by scoring whole items by teacher forcing over the entire vocabulary. Cells are discarded when the field holds fewer than three items, when fewer than two survive the restriction, or when the calibration residual exceeds 0.05; mass appearing at the restricted prompt on items absent from the unrestricted field is outside both models' support and is excluded from the renormalization rather than smoothed.

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